

Feasibility Analysis & MVP Roadmap

Project: Climate Response Hub

Team: Program Counter

1. Executive Summary

Currently, emergency coordinators at Toronto Emergency Management (TEM) and Toronto Shelter & Support Services (TSSS) face significant cognitive load during extreme heatwaves and outage-related climate events, as they must mentally connect fragmented data across multiple tabs.

Climate Response Hub is a decision-support dashboard that integrates climate risk data, vulnerability indicators, and shelter access gaps into a single, unified view. Acting as a planning and coordination layer, it enables coordinators to log their dispatch actions and track real-time status. Importantly, the platform does not directly dispatch emergency services or control infrastructure, keeping the operational scope focused, realistic, and highly feasible for municipal adoption.

2. Feasibility Analysis

This solution meets the practical business and technical feasibility criteria required by the hackathon, directly addressing the goals of municipal and utility partners.

A. Operational Feasibility

- **Focused Planning & Coordination:** The dashboard fits seamlessly into existing workflows by making risk and resources easier to understand without replacing the human decision-maker. It prevents duplicate efforts: when a coordinator logs a specific action (e.g., requesting 20 mobile cooling units for Mitchell Field Community Center in Willowdale East), the neighborhood is marked, and others can update the status to 'Deployed' once the units arrive.
- **Reduced Decision-Making Time:** By eliminating the "tab-switching" cognitive load, the platform accelerates crisis triage, transforming hours of manual data cross-referencing into seconds of actionable insights.

B. Economic Feasibility

- **Resource Optimization:** Efficiently targets limited physical relief budgets and resources to high-risk areas with the highest vulnerability scores (e.g., a score of 81) and severe accessibility gaps.
- **Cost-Effective Architecture:** The core infrastructure costs are extremely low. It leverages existing public data ecosystems (e.g., the City of Toronto's Heat Relief Network) and Esri's infrastructure, requiring very low initial capital for deployment.

C. Technical Feasibility

- **Lightweight & Scalable MVP:** For rapid prototyping and reliable performance, the current MVP uses React's state management and localStorage to simulate a real-time collaboration database without the overhead of maintaining heavy backend servers.
- **Rule-based Triage & Advanced Computing:** Employs rule-based triage logic alongside the Open-Meteo API (12-hour forecasts) and the ArcGIS Spatial Analysis API (distance calculations) to actively identify localized high-risk hotspots.
- **Clear MVP Boundaries (Real + Mock Data):** By explicitly excluding complex integrations like live dispatch systems or real-time grid control for the MVP, the technical scope remains highly achievable. The platform easily handles real-time infrastructure status by using Mock APIs for cooling center capacity (e.g., 70% full) and backup power visibility, designed for easy swapping with live utility APIs in the future.

3. MVP Scope & 6-Month Roadmap

Instead of overbuilding, our roadmap focuses on a minimum viable product that immediately solves the core collaboration problem, followed by strategic expansions.

- **Phase 1: Current MVP (Months 0-1)**
 - **Core Use Case 1 (Identify Risk):** Integration of Esri GIS Heat Vulnerability Index and rule-based triage logic to calculate exact distances to the nearest cooling centers.
 - **Core Use Case 2 (Log Action):** Implementation of the collaborative CRUD workflow using React localStorage (e.g., Track Status: Pending → Deployed).
- **Phase 2: Data & Partner Integration (Months 2-3)**
 - Transition from localStorage to a robust database (e.g., MongoDB) with strict Role-Based Access Control (RBAC) to securely handle sensitive emergency

operations data. No individual resident data will be stored, ensuring strict privacy.

- Transition from mock infrastructure data to actual utility APIs (e.g., Alectra Utilities) to monitor live power grid and backup generator statuses during extreme weather events.
- Phase 3: Service Expansion (Months 4-6)
 - Expansion beyond heatwaves to address winter cold snaps and early warning modules for other climate risks such as severe snowstorms and flash floods.
 - Development of automated Energy Equity reporting modules to assist utility partners with long-term infrastructure upgrade planning.

4. Strategic Impact

The Climate Response Hub goes beyond a simple monitoring tool; it serves as a critical infrastructure piece to achieve national and corporate resilience targets.

- Alignment with Government of Canada Goals: Directly supports the national target ensuring that by 2026, 80% of health regions will have implemented evidence-based adaptation measures, and strictly aligns with the ultimate goal to eliminate extreme heatwave deaths by 2040.
- Supporting Alectra Utilities: By providing unprecedented visibility into backup power and community vulnerability, the platform actively supports Alectra's climate resilience and emergency preparedness goals.